

EXPONENTIAL FUNCTIONS (Chapter 6)

LESSON 1 - Characteristics of Exponential Functions

Exponential functions are equations where one of the variables is in the exponent, and can be written in the form $y = a(b)^x$. Use your graphing calculator to sketch the following functions and complete the chart below.

Exponential Function	Graph	x-intercept	$a = \underline{\hspace{1cm}}$	y-intercept	$b > 1$ or $0 < b < 1$	End Behaviour	Domain Range
$f(x) = 2^x$							
$f(x) = 3(2)^x$							
$f(x) = 3\left(\frac{1}{2}\right)^x$							
$f(x) = 2\left(\frac{1}{3}\right)^x$							

1. What characteristic of the graph is associated with the a value of the function?
2. How does the b value affect the shape of the graph?
3. Do any of the graphs cross the x -axis?
4. What would happen if $b = 1$? Try graphing $y = 1^x$.

Summary

The a value determines the _____ of the graph.

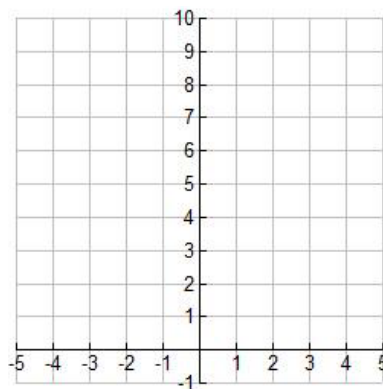
The b value determines the _____ of the graph.

If $b > 1$, the graph is _____.

If _____, the graph is decreasing.

On the graph to the right sketch $y = 2(2)^x$, $y = 2(3)^x$ and $y = 2(5)^x$.

From this we see that the larger the b value the _____ the function increases.



The domain of all exponential graphs is _____.

The range of all exponential graphs of the form $y = a(b)^x$ is _____.

In Summary

Key Ideas

- An exponential function has the form $f(x) = a(b)^x$, where x is the exponent and $a \neq 0$, $b > 0$, and $b \neq 1$.
- All exponential functions of the form $f(x) = a(b)^x$, where $a > 0$, $b > 0$, and $b \neq 1$, have the following characteristics:

Number of x-Intercepts	0
y-Intercept	a
End Behaviour	Curve extends from quadrant II to quadrant I.
Domain	$\{x \mid x \in \mathbb{R}\}$
Range	$\{y \mid y > 0, y \in \mathbb{R}\}$

Need to Know

- There are two different shapes of the graphs of an exponential function of the form $f(x) = a(b)^x$, where $a > 0$, $b > 0$, and $b \neq 1$:

- Case 1: An increasing function; the curve extends from quadrant II to quadrant I.

Increasing



Decreasing



- Case 2: A decreasing function; the curve extends from quadrant II to quadrant I.

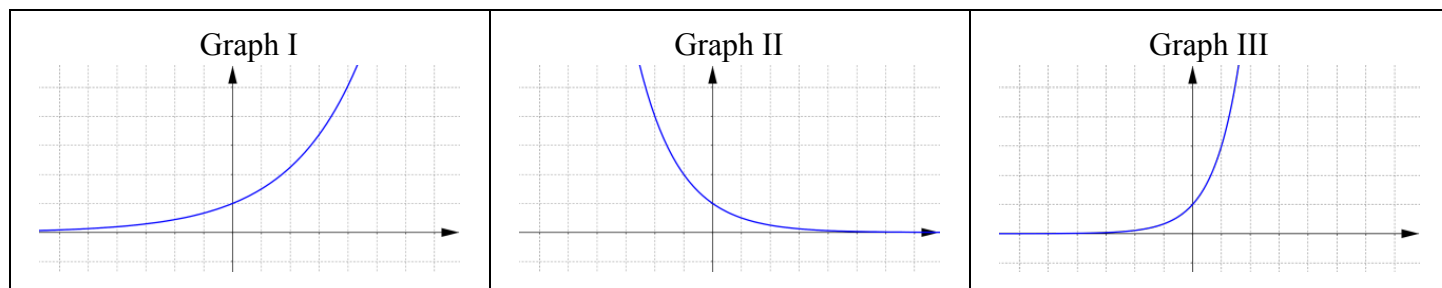
Practice

1. Answer the following questions **without using your graphing calculator**.

$y = 2(7)^x$	$y = 3\left(\frac{1}{4}\right)^x$	$y = 6(0.8)^x$
x-intercept:	x-intercept:	x-intercept:
y-intercept:	y-intercept:	y-intercept:
Increasing or decreasing?	Increasing or decreasing?	Increasing or decreasing?
Domain:	Domain:	Domain:
Range:	Range:	Range:

2. Verify your answers in #1 by graphing each function on your calculator.

3. Using the following graphs to answer the questions below.



*Note: All of the above graphs have a scale of 1 unit for both the x and y axis.

- The graph(s) that is/are increasing is/are _____.
- The graph with the largest b value is _____.
- The graph(s) with $0 < b < 1$ is/are _____.
- The a value for all three graphs is _____.

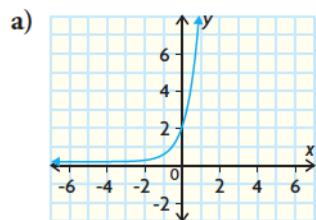
4. Examine each function to determine if it increases or decreases. Then match each function with the corresponding graph below. **Justify** your reasoning.

I. $y = 3(0.2)^x$

II. $y = 4(3)^x$

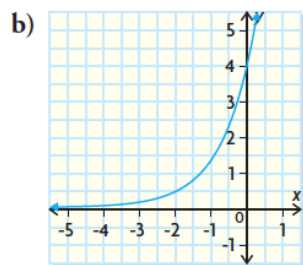
III. $y = 4(0.5)^x$

IV. $y = 2(4)^x$



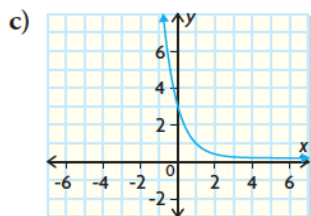
Equation: _____

Justification:



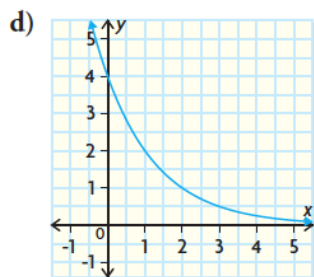
Equation: _____

Justification:



Equation: _____

Justification:



Equation: _____

Justification:

LESSON 2 - Exponential Regression & Applications of Exponential Functions

VOCABULARY

Exponential growth function:

- An exponential function whose y -values increase as you move from left to right along the x -axis
- for an exponential function of the form $y = a(b)^x$, exponential growth occurs when $a > 0$ and $b > 1$.

Exponential decay function:

- An exponential function whose y -values decrease as you move from left to right along the x -axis
- for an exponential function of the form $y = a(b)^x$, exponential decay occurs when $a > 0$ and $0 < b < 1$.

Half-life exponential function: A function of the form $y = a\left(\frac{1}{2}\right)^{\frac{x}{h}}$ where the base is $\frac{1}{2}$, $a \neq 0$, and $h \neq 0$; the value of h called the half-life because it corresponds to the point on the graph of the function where the value of the function is half its initial value, a .

Future value: The amount that an investment will be worth after a specified period of time.

Principal: The original amount of money that is invested or borrowed.

Compounding period: The time over which interest is calculated and paid on an investment or loan.

Depreciation: The decrease in the value of an asset over time.

Appreciation: The increase in the value of an asset over time.

Just like with polynomials, we need to be able to determine the equation of an exponential function by performing a regression.

Recall: Exponential functions are of the form:

$$y = a(b)^x$$

EXAMPLE 1

1. The population growth of two different bacteria was observed and the results are shown below.

Bacteria A

Time (hrs)	0	1	2	3	4	5
# of Bacteria	4	8	16	32	64	128

Bacteria B

Time (hrs)	0	1	2	3	4	5
# of Bacteria	2	6	18	54	162	486

	Bacteria A	Bacteria B
a) Exponential regression function.		
b) What do you notice about the a value?		
c) At what rate is each bacteria growing each hour? What percentage is that?		
d) State the b value.		
e) What do you notice about the b value and the rate at which the populations were growing?		
f) What would a b value of 1 mean in these situations?		

EXAMPLE 2

1. Examine the two sets of data below and fill in the accompanying table.

Value of Car

Time (Years)	0	1	2	3	4
Value of Car (\$)	22 500	18 900	15 876	13335.84	11202.11

Value of Calculator

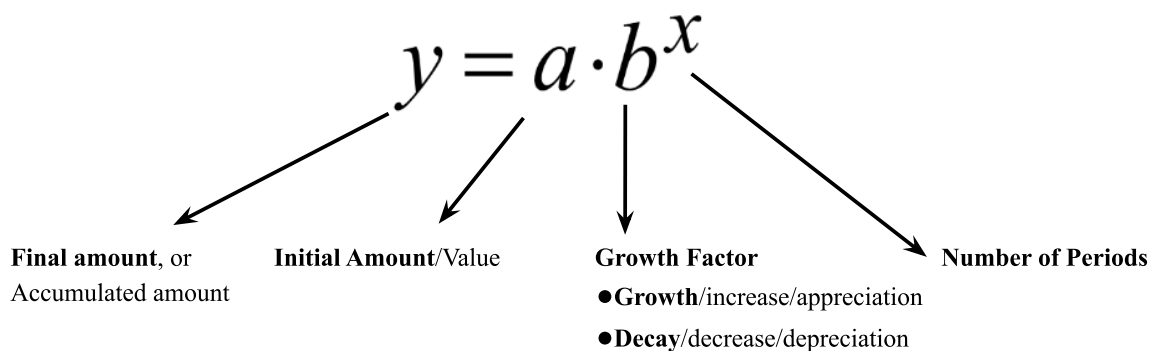
Time (Years)	0	1	2	3	4
Value of Calculator (\$)	125	118.75	112.81	107.17	101.81

	Car	Calculator
a) Exponential regression function.		
b) What do you notice about the a value?		
c) By what percentage is the value of each item decreasing each year?		
d) State the b value.		
e) What do you notice about the b value and the rate at which the value is decreasing?		
f) If the b value was larger than 1 what would be happening to the value of these items?		

Summary

In the equation $y = a(b)^x$:

- The a value is the y -intercept of the graph and the _____ of the data.
- The b value represents the growth factor.
- $b > 1$: If the data is increasing, the b value must be _____ than one. For example,
 - a b value of _____ indicates the data is doubling each time period.
 - a b value of 1.15 indicates the data is increasing by _____% each time period.
- $0 < b < 1$: A b value between zero and one indicates the data is _____. In these cases, the b value indicates the amount of data that is continuing in the pattern. For example:
 - a b value of 0.75 indicates 75% of the data continues in the pattern, or that there is a _____% decrease each time period.



EXAMPLE 3

Given the following exponential functions, determine the starting value and the growth factor.

Function	Starting Value	Growth Factor	Increase or Decrease
$y = 100(0.67)^x$			
$y = 90(2.5)^x$			

Practice

1. Laurie invested \$2 500 in an account that compounds monthly. The following table gives the value of her investment for the first five months.

Time (months)	0	1	2	3	4	5
Value of Investment (\$)	2500.00	2507.29	2514.60	2521.94	2529.29	2536.67

a) Use exponential regression to **determine** the compound interest function that models this situation. Round the a -value to the nearest whole and the b value to nearest thousandth.

b) What is the rate of interest, to the nearest tenth of a percent, that Laurie earns per month?

c) How long, in months, will it take for Laurie's investment to grow to \$3 000?

2. If a basketball is properly inflated, it bounces to 75% of its original height. A basketball is dropped from a height of 5 m.

a) Complete the table:

Bounce	0	1	2	3	4	5
Height (m)	5					

b) State the regression equation that best models this bouncing ball. Round the a -value to the nearest whole and the b value to nearest hundredth.

c) After how many bounces does the ball rebound to less than 0.3 m?

3. The population of Canada from beginning in 1871 is shown in the table below.

Years since 1871	0	10	20	30	40	50	60	70	80
Pop of Canada (millions)	2.40	3.23	3.74	5.42	7.22	8.80	10.38	11.51	14.01

a) State the exponential regression function that represents the data. Round values to the nearest thousandth. State the growth rate, to the nearest hundredth of a percent.

b) Assuming that population growth continued at the same rate, estimate the population in 2011.

c) In which year will Canada reach of population of 50 million?

4. The function that models the decay of the element Mathonium is

$$A(t) = 100\left(\frac{1}{2}\right)^{\frac{t}{5730}}$$

where $A(t)$ represents the amount of Mathonium in grams remaining, and t is the time in years.

a) **Determine** how much Mathonium, to the nearest hundredth, that will be present in 10 000 years.

b) How many years, to the nearest year, will it take for there to be only 20 grams of Mathonium left?

5. Darren bought a camera two years ago for \$2 000 and has been told that it will depreciate at a rate of 20% per year.

a) **Determine** an exponential equation that would represent the situation.

b) What will the camera be worth in 5 years?

c) How many years, to the nearest year, until the camera is worth \$200?

6. The number of insects in a colony doubles every month. If there are now 250 insects, **determine** how long will it take for the colony to grow to 512 000.

7. A radioactive element has a half-life of one week. **Determine** the percent of the element that will remain after 4 weeks, to the nearest hundredth or a percent.

8. Tanya invested in a GIC. The following table illustrates the growth of the investment over 5 years.

End of Year	Amount (\$)
1	628.50
2	658.35
3	689.63
4	722.38
5	756.70

- a) How much did Tanya initially invest?
- b) What is the average rate of return (growth rate), to the nearest hundredth of a percent?
- c) Predict the value of the GIC at the end of 10 years.
- d) How many years will it take for the GIC to be worth \$1100?

LESSON 3 - Review of Exponent Laws

$$x^m$$

Label the following using the words *power*, *base*, and *exponent*.

Exponent Law		Example
Product Law	$x^m \times x^n = x^{m+n}$	$x^3 \times x^4 = x^{3+4}$ $= x^7$
Quotient Law	$x^m \div x^n = \frac{x^m}{x^n} = x^{m-n}$	$\frac{x^3}{x^{-2}} = x^{3-(-2)}$ $= x^5$
Power of a Power Law	$(x^m)^n = x^{m \cdot n}$	$(x^5)^4 = x^{5 \cdot 4}$ $= x^{20}$
Power of a Product Law	$(xy)^m = x^m y^m$	$(x^2 y^3)^4 = x^{2 \cdot 4} y^{3 \cdot 4}$ $= x^8 y^{12}$
Power of a Quotient Law	$\left(\frac{x}{y}\right)^m, y \neq 0$	$\left(\frac{2x^3}{y^4}\right)^2 = \frac{2^2 x^{3 \cdot 2}}{y^{4 \cdot 2}}$ $= \frac{4x^6}{y^8}$
Zero Exponent Law	$x^0 = 1$	$3x^0 = 3 \cdot 1$ $= 3$
Negative Exponent	$x^{-n} = \frac{1}{x^n}$ or $\frac{1}{x^{-n}} = x^n$	$3^{-2} = \frac{1}{3^2}$ $= \frac{1}{9}$
Rational Exponent	$x^{\frac{m}{n}} = (\sqrt[n]{x})^m$ or $(\sqrt[n]{x^m})$	$8^{\frac{2}{3}} = (\sqrt[3]{8})^2$ $= (2)^2$ $= 4$

Practice

1. Write each expression without brackets and with positive exponents.

a) $4xy^{-3}$

b) $\frac{15y^3}{5y}$

c) $(3x^3y)(5x^{-2}y^4)$

d) $\frac{24x^{-8}}{16x^{-3}}$

e) $\frac{2}{y^{-\frac{1}{3}}}$

f) $(2x^{-2})^3$

2. Simplify each of the following. Write the simplified expressions with positive exponents.

a) $\frac{x^5y^{-1}}{x^2y^{-4}}$

b) $\left(\frac{5x^3}{2y^4}\right)^{-3}$

c) $(4m^2n)^{-1} \times 2mn^5$

d) $\frac{3x^2y^0z^{-4}}{(2xyz)^3}$

3. Simplify, leave answers written as a power.

a) $4^3(4)^2$

b) $(5)^7 \cdot (5)^{-2}$

c) $(6)^2 \div (6)^{-3}$

d) 2^{-4} (as a fraction)

e) $(3^2)^4$

4. Without using a calculator, determine the exact value of the following.

a) 5^{-2}

b) $27^{\frac{4}{3}}$

c) $\left(\frac{4}{9}\right)^{-\frac{3}{2}}$

d) $125^{\frac{1}{3}} - 10^0(64)^{\frac{2}{3}}$

Changing Base

EXAMPLE

Convert each of the following to the base indicated.

a) 8 as a base of 2

b) 9^{5x} as a base of 3

c) $\frac{1}{64}$ as a base of 2

d) 125^{2-x} as a base of 5

e) $\left(\frac{27}{64}\right)^{x+2}$ as a base of $\frac{3}{4}$

f) $8 \cdot 16^x$ as a base of 2

Practice

Convert each of the following to the base indicated.

5. a) 32^x as a base of 2

b) 81^{x-2} as a base of 3

c) $\frac{1}{27^{2x}}$ as a base of 3

d) $\left(\frac{1}{16}\right)^{x+1}$ as a base of 2

e) $\left(\frac{25}{49}\right)^{3x}$ as a base of $\frac{5}{7}$

f) $2 \cdot 4^x$ as a base 2

g) $9 \cdot 27^{x-1}$ as a base 3

h) $\frac{1}{4} \cdot \left(\frac{1}{16}\right)^{4-x}$ as a base 2

LESSON 4 - Solving Exponential Equations

VOCABULARY

Exponential equation: is an equation where the variable, x , is in the exponent.

Example: $2^x = 8$

Solving Exponential Equations with common base

We will have two ways to solve our equations: **Algebraically & Graphically**

Strategies for Solving Exponential Equations With a Common Base

(Type#1) If the bases are the same:

- Ignore the bases and set the exponents equal to each other.
- Solve for x .

$$\begin{aligned} 11^{2x+6} &= 11^2 \\ 2x + 6 &= 2 \\ 2x &= -4 \\ x &= -2 \end{aligned}$$

(Type#2) If the bases are not the same, but can be written such that both sides have the same base.

- Rewrite the bases on each side of the equation to a common base.
- Substitute the rewritten bases into the original equation.
- Simplify the exponents.
- Drop the bases.
- Solve for x .

$$\begin{aligned} 4^{2x} &= 8^{x+1} \\ (2^2)^{2x} &= (2^3)^{x+1} \\ 2^{4x} &= 2^{3x+3} \\ 4x &= 3x + 3 \\ x &= 3 \end{aligned}$$

Strategies for Solving Exponential Equations WITHOUT a Common Base

(Type#3) If the bases are not the same, and they cannot be written such that both sides have the same base, you can use Trial & Error (guess and check).

Trial & Error

- Use an educated guess and check method to estimate the answer.

$$1.4^x = 7$$

$$\text{Guess 1: } x = 5; 1.4^5 = 5.37824.. < 7$$

$$\text{Guess 2: } x = 7; 1.4^7 = 10.54135... > 7$$

$$\text{Guess 3: } x = 6; 1.4^6 = 7.529536... \approx 7$$

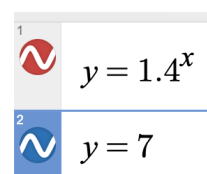
So, x is approximately equal to 6.

****Note:** You should always include the guess that comes before and after the actual answer.

(Type#4) Graphing Calculator

Finding the x -coordinate of the intersection point

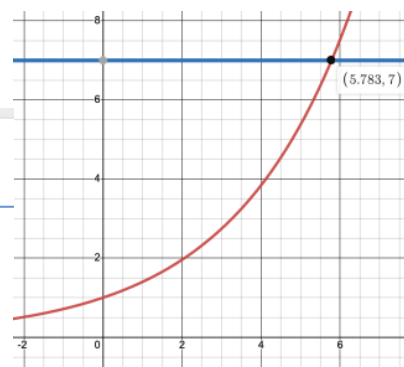
- 1) Y1=left side of the equation
Y2=right side of the equation
- 2) x : [$xmin$, $xmax$, $xsc1$]
 y : [$ymin$, $ymax$, $yscl$]
- 3) calc INTERSECT
- 4) Solution(s)



1 $y = 1.4^x$

2 $y = 7$

$x = 5.783..$



EXAMPLE 1

Solve the following exponential equations algebraically.

a) $5^{2x+3} = 5^7$

b) $7^{x-2} = 7^{2x-3}$

c) $4^{8x} = \frac{1}{4}$

EXAMPLE 2

Solve the following exponential equations graphically.

a) $5^{2x+3} = 5^7$

b) $7^{x-2} = 7^{2x-3}$

c) $4^{8x} = \frac{1}{4}$

EXAMPLE 3

Solve the following exponential equations algebraically.

a) $9^{x-1} = 3^{x-3}$

b) $2(4)^{2x} = 32$

c) $5^{x+2} = \sqrt{125}$

EXAMPLE 4

Solve the following exponential equations graphically.

a) $2^x = 7$

b) $3(5)^x = 10$

c) $4^{3x+1} = 9$

Practice

1. **Algebraically determine** the solution to the following exponential equations.

a) $2^{2-3x} = 2^{7x-8}$	b) $9^{x+1} = 729$
c) $5^{4x+2} = 25^x$	d) $2^{-x} = 64$
e) $4^{2x} = \frac{1}{32}$	f) $8^{x-1} = 128^{3x+1}$

g) $5(16)^{3x-5} = 10$	h) $\left(\frac{1}{25}\right)^{x+1} = 125$
i) $256\left(\frac{1}{2}\right)^x = 512$	j) $2^x = 16\sqrt{2}$ *challenge

2. Consider the exponential equation $\left(\frac{4}{7}\right)^{5x} = \left(\frac{64}{343}\right)^{2x-1}$.

a) Solve the equation using a graphical method on a calculator.

b) Solve the equation algebraically by converting each side to a common base.

3. The number of solutions to the equation $16(8)^{x-1} = 2^x$ is

- a. 0
- b. 1
- c. 2
- d. None of the above

Use the following information to answer the next question.

Consider the exponential equation $25^{x+1} = 5^{3(x-1)}$.

4. The solution to the above equation, to the nearest tenth, is _____.

Use the following information to answer the next question.

Consider the exponential equation $8^{2x-1} = 16$.

5. The solution to the above equation, to the nearest tenth, is _____.

6. A culture of 2 000 bacteria that doubles in size every 6 days is represented by the function

$$P(t) = 2000(2)^{\frac{t}{6}}$$

where $P(t)$ represents the number of bacteria and t represents time in days.

a) **Determine** how many bacteria will be present in 25 days.

b) **Determine** how many days it will take until the bacteria reaches a population of 16 000.

7. A radioactive isotope has a mass of 512 grams and a half life of 45 minutes. The number of grams, N , present after t minutes is given by the formula $N(t) = 512\left(\frac{1}{2}\right)^{\frac{t}{45}}$. **Determine** the time it takes for the mass to reduce to 1 gram.